

1. AIM

To design an CMOS inverter repeater using minimum channel length and minimum width in 180nm technology and determine

- 1] R_{out} , (output resistance of the inverter)
- 2] C_{out} , (output capacitance of the inverter)
- 3] intrinsic delay t_p , (intrinsic delay of the inverter)
- 4] R_n , R_p (resistance of NMOS and PMOS)
- 5] k (number of stage of repeaters)

to calculate propagation delay without inverter/buffer insertion and with inverter/buffer insertion and to compare the results.

2. TOOL

Cadence Virtuoso

3. THEORY

1. Unbuffered Interconnect Delay: A long interconnect exhibits distributed resistance R_{int} and capacitance C_{int} , causing propagation delay to scale quadratically with wire length. Driven by a minimum-sized inverter (R_{out} , C_{out}), the total unbuffered delay is given by $T_{50\%} = 0.7R_{out}(C_{int} + C_{out}) + R_{int}(0.4C_{int} + 0.7C_{out})$.

2. Minimum-Sized Repeater Insertion: Inserting intermediate minimum-sized inverters divides the line into k shorter sections, converting quadratic delay growth into a linear relationship. The total propagation delay for k minimum-sized repeater stages is given by $T_{50\%} = k \left[0.7R_{out} \left(\frac{C_{int}}{k} + C_{out} \right) + \frac{R_{int}}{k} \left(0.4 \frac{C_{int}}{k} + 0.7C_{out} \right) \right]$.

3. Optimum Number of Stages: Minimizing the total propagation delay with respect to k yields the optimum number of repeater stages: $k = \sqrt{\frac{0.4R_{int}C_{int}}{0.7R_{out}C_{out}}}$.

4. TABLES

Table 4.1: Design Parameters

given parameter	value	unit
technology	180	nm
V_{dd}	1.8	V
V_{tn}	0.45	V
$ V_{tp} $	0.45	V
minimum width of nmos	400	nm
minimum width of pmos	400	nm
k_n'	70	$\mu\text{A}/\text{V}^2$
k_p'	20	$\mu\text{A}/\text{V}^2$
R_{int}	68	$\text{k}\Omega$
C_{int}	60	fF

Table 4.2: Derived and observed intrinsic parameters

derived and observed parameter	value	unit
t_{phl}	9.8587	ps
t_{plh}	27.33919	ps
t_p	18.5989	ps
R_n	4.762	$\text{k}\Omega$
R_p	16.67	$\text{k}\Omega$
R_{out}	10.716	$\text{k}\Omega$
C_n	3.0004	fF
C_p	2.3768	fF
C_{out}	3.0004	fF
k (number of repeater stages)	8	unitless

Table 4.3: Result

parameter	without inv/buf insertion	with inv/buf insertion	unit	error
t_{phl}	2.882	1.17325	ns	-59.2904%
t_{plh}	3.119	1.23534	ns	-60.3931%
t_p	3.0005	1.2043	ns	-59.8634%

Table 4.4: comparison of theoretical and practical values

parameter	theoretical value (T_p)	practical value (t_p)	unit	signed error
without inv/buf insertion	2.2473	3.0005	ns	+33.5158%
with inv/buf insertion	0.9768	1.2043	ns	+23.2903%

5. CALCULATION

$$\begin{aligned}
 R_n &= 1000/(k_n * (W/L) * (V_{dd} - V_{tn})) \\
 &= 1000/(70 * (400/180) * (1.8 - 0.45)) \\
 &= 4.7619048k\Omega \approx 4.762k\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_p &= 1000/(k_p * (W/L) * (V_{dd} - V_{tp})) \\
 &= 1000/(20 * (400/180) * (1.8 - 0.45)) \\
 &= 16.6666667k\Omega \approx 16.67k\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_{out} &= (R_n + R_p)/2 \\
 &= (4.7619048 + 16.6666667)/2 \\
 &= 10.7142858k\Omega \approx 10.714k\Omega
 \end{aligned}$$

$$\begin{aligned}
 C_{out1} &= t_{phl}/(0.69 * R_n) \\
 &= 9.8587/(0.69 * 4.7619048) \\
 &= 3.0004739ff \approx 3ff
 \end{aligned}$$

$$\begin{aligned}
 C_{out2} &= t_{plh}/(0.69 * R_p) \\
 &= 27.33919/(0.69 * 16.6666667) \\
 &= 2.3773209ff \approx 2.3773ff
 \end{aligned}$$

$$\begin{aligned}
 C_{out} &= \max(C_{out1}, C_{out2}) \\
 &= \max(3.0004739, 2.3773209) \\
 &= 3.0004739ff \approx 3ff
 \end{aligned}$$

$$\begin{aligned}
 t_{p(min)} &= (t_{phl} + t_{plh})/2 \\
 &= (9.8587 + 27.33919)/2
 \end{aligned}$$

$$= 18.598945ps \approx 18.5989ps$$

$$k = \sqrt{(R_{int} * C_{int} * 0.4) / (R_{out} * C_{out} * 0.7)}$$

$$= \sqrt{(68 * 60 * 0.4) / (10.7142858 * 3.0004739 * 0.7)}$$

$$= 8.5159777 \approx 8$$

$$T_{p(no_rep)(theoretical)} = (0.7 * R_{out} * (C_{int} + C_{out})$$

$$+ R_{int} * (0.4 * C_{int} + 0.7 * C_{out})) / 1000$$

$$= (0.7 * 10.7142858 * (60 + 3.0004739)$$

$$+ 68 * (0.4 * 60 + 0.7 * 3.0004739)) / 1000$$

$$= 2.2473261ns \approx 2.2473ns$$

$$T_{p(rep)(theoretical)} = (k * (0.7 * R_{out} * (C_{int}/k + C_{out})$$

$$+ (R_{int}/k) * (0.4 * C_{int}/k + 0.7 * C_{out}))) / 1000$$

$$= (8 * (0.7 * 10.7142858 * (60/8 + 3.0004739)$$

$$+ (68/8) * (0.4 * 60/8 + 0.7 * 3.0004739))) / 1000$$

$$= 0.976851ns \approx 0.9769ns$$

$$t_{HL(no_rep)} = 2.882ns$$

$$t_{LH(no_rep)} = 3.119ns$$

$$t_{p(no_rep)} = (t_{phl_no_rep} + t_{plh_no_rep}) / 2$$

$$= (2.882 + 3.119) / 2$$

$$= 3.0005ns$$

$$t_{HL(rep)} = 1.17325ns$$

$$t_{LH(rep)} = 1.23534ns$$

$$t_{p(rep)} = (t_{phl_rep} + t_{plh_rep}) / 2$$

$$= (1.17325 + 1.23534) / 2$$

$$= 1.204295ns \approx 1.2043ns$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

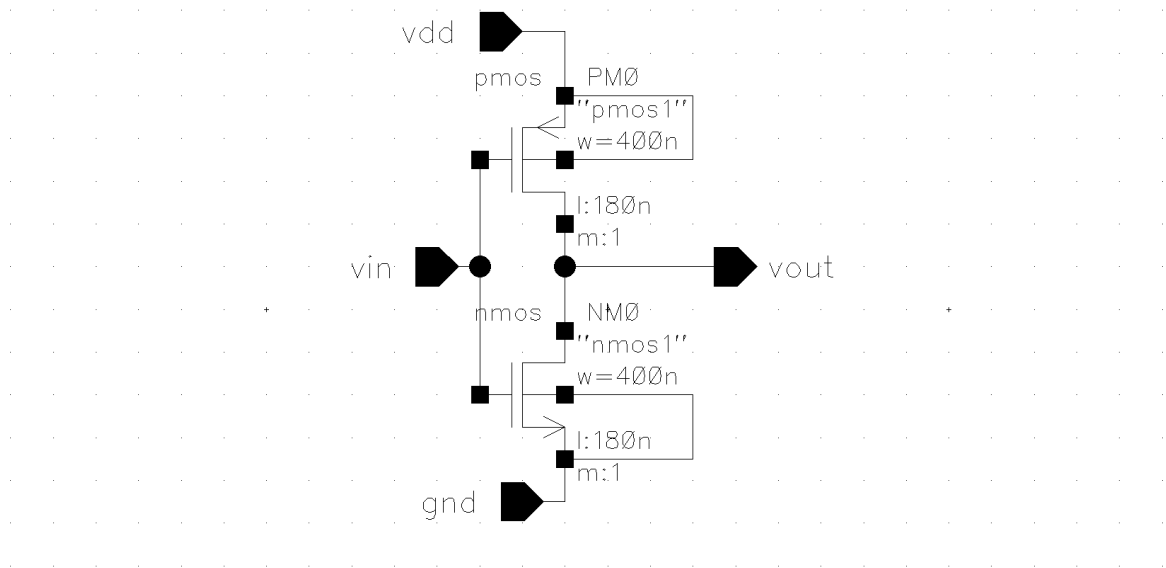


Figure 6.1: Schematic of minimum size inverter

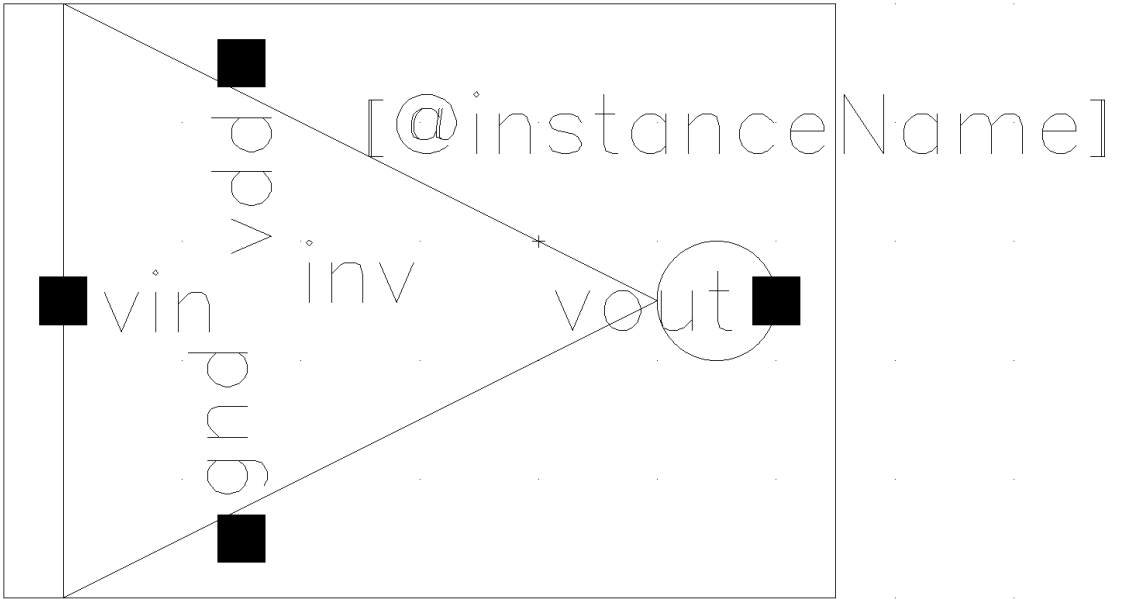


Figure 6.2: Symbol of minimum size inverter

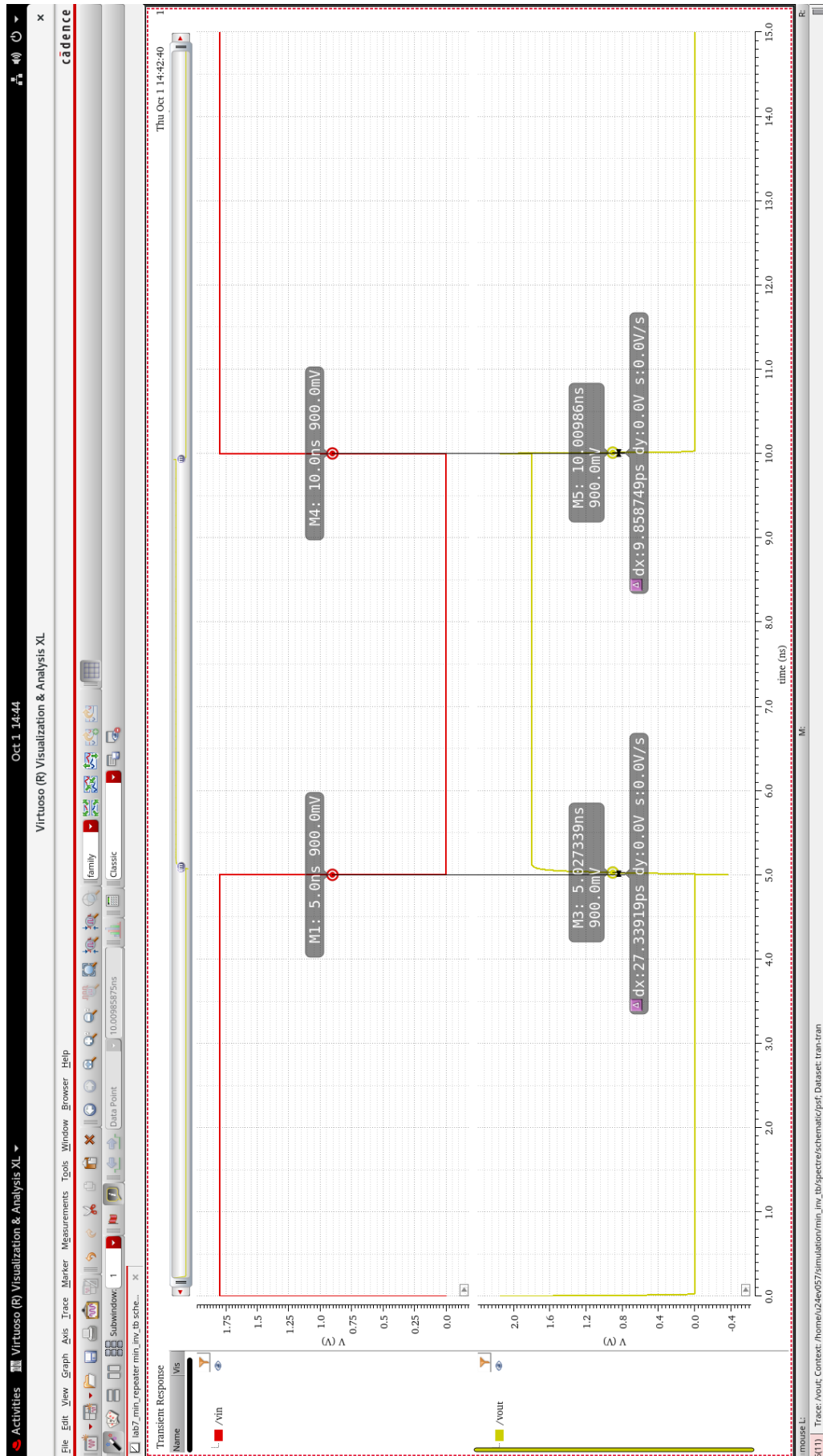


Figure 6.3: Transient waveform of minimum size inverter without load

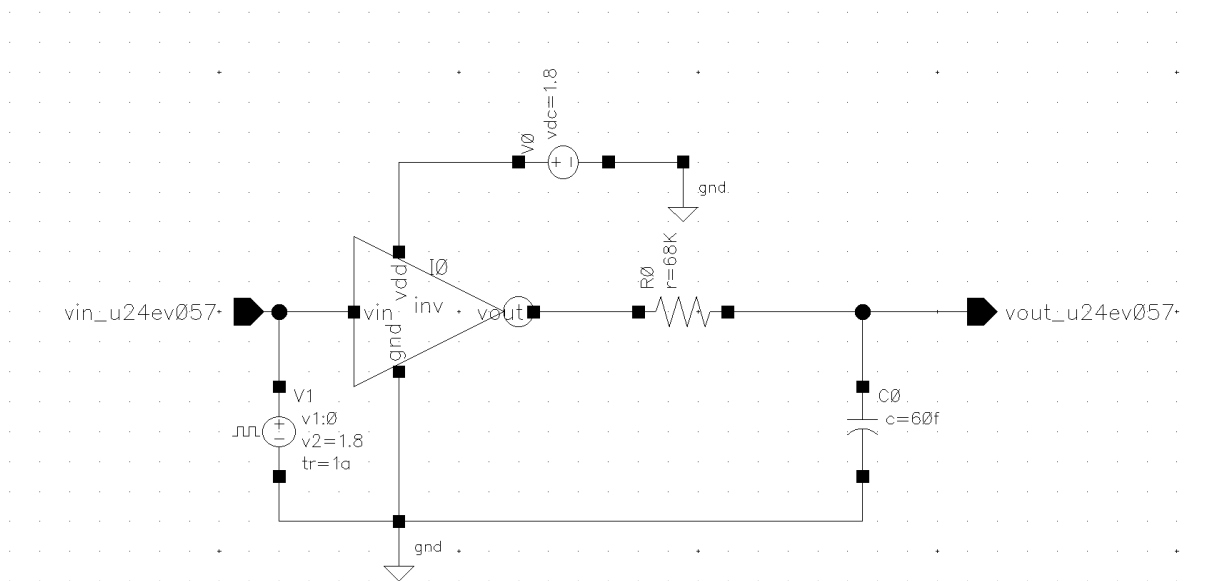


Figure 6.4: Schematic of inverter without inv/buf insertion (without repeaters)

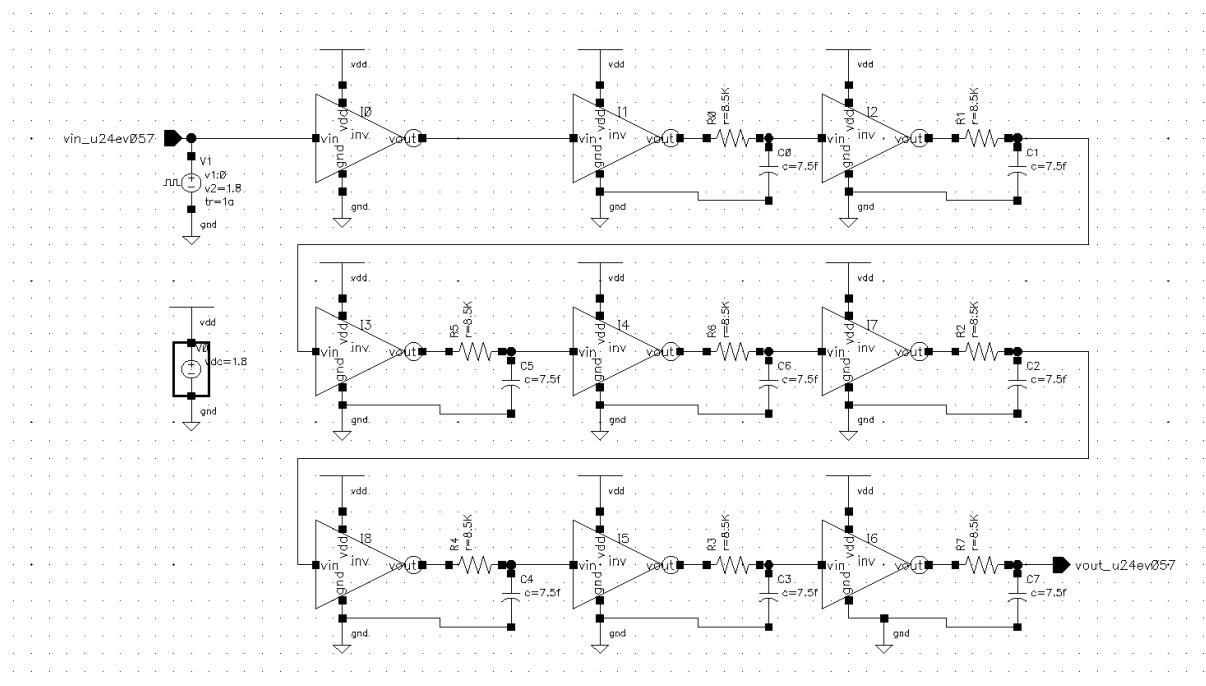


Figure 6.5: Schematic of inverter with inv/buf insertion (with repeaters)

7. CONCLUSION

1. An unbuffered interconnect and minimum-sized CMOS inverter repeaters were designed and simulated in 180 nm technology ($V_{DD} = 1.8\text{V}$, $R_{int} = 68\text{k}\Omega$, $C_{int} = 60\text{fF}$).
2. Minimum-sized repeater insertion ($k = 8$) reduced propagation delay from 3.0005ns down to 1.2043ns (59.86% reduction), in good agreement with the theoretical delay of 0.9768ns .